portto: Offering a highly accessible and reliable crypto wallet with cloud infrastructure

Tell us your challenge. We're here to help.

Contact us (/contact)

About portto

portto deploys its wallet solution or Cloud to provide user experiences high traffic surge guaranteeing dat for blockchain tra

About portto

Founded in 2019, portto is a blockchain tech company based in Taiwan dedicated to developing and offering more accessible blockchain solutions to enterprises and the general public. Blocto, its all-in-one cross-chain crypto wallet, provides a simple way to manage cryptocurrencies, dApps, and NFTs to 1.6 million users around the world. portto has also partnered with a dozen of industry leaders like Moto GP Ignition and NBA to launch blockchain-based services.

Google Cloud results

- Shortens file access latency through global storage supported by Cloud Storage
- Enables smooth and cost-effective blockchain data indexing with Cloud Bigtable
- Supports highly secure key management for crypto transactions with Cloud Key Management

Seam
suppc
instar
surge

Industries: Technology

Location: Taiwan

About CloudMile

As one of the leading AI and cloud service providers in Taiwan, CloudMile empowers businesses to accelerate digital transformation through cloud technology and machine learning. Holding more than 130 professional certificates, it currently offers services to 700+ customers across Singapore, Malaysia, Taiwan, and Hong Kong. In 2020, CloudMile became a Google Cloud Managed Service Provider, including 40 from Google Cloud.

- Google Cloud (https://cloud.google.com/)
- Cloud Bigtable (https://cloud.google.com/bigtable/)
- Cloud SQL (https://cloud.google.com/sql/)
- Cloud Key Management Service (https://cloud.google.com/kms/)
- Cloud Storage (https://cloud.google.com/storage/)
- Compute Engine (https://cloud.google.com/compute/)

The blockchain technology has
applied to various domains over

From cryptocurrency, decentrali
(dApps) to non-fungible tokens (
has been a major driving force b
development of Web 3.0, a dece
the internet.

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud IAM (https://cloud.google.com/iam/)

However, leveraging decentralized networks often requires a high level of tech savviness and complex procedures. For example, to own a crypto wallet for blockchain transactions, users normally need to backup a seed phrase of 12-14 words used to recover the wallet, open an account on a cryptocurrency exchange platform to top up the wallet and pay transaction fees, and acquire some basic knowledge about blockchain to be able to track their assets and transaction records. For some, this could be daunting and prevent them from taking advantage of the blockchain technology.

portto (https://portto.com/) is committed to making blockchain more accessible to the general public. Founded in 2019, the Taiwan-based tech company offers easy-to-use blockchain solutions to enterprises and individual users. Blocto, its all-in-one cross-chain crypto wallet, enables users to create a crypto wallet simply with an email address or a social media account within a few seconds. Through its partnership with industry leaders like Moto GP Ignition and NBA, portto has been offering Blocto (https://blocto.io/) to support a number of blockchain-based services and quickly gained users worldwide. Currently, around 1.6 million people around the world are using Blocto to manage their cryptocurrencies, dApps, and NFTs.

"To ensure smooth user experiences and leverage resources in the most cost-effective way, we needed to run our crypto wallet with a cloud infrastructure that can be easily scaled up and down at any time. Google Cloud offers excellent infrastructure scalability with proven high security, which is also essential for our services."

—Hsuan Lee, Co-Founder and CEO, portto

"Our core mission is to bring the benefits of blockchain to more people. That's why we have been developing the most user-friendly way for developers and crypto wallet users to leverage the blockchain technology," explains Hsuan Lee, Co-founder and CEO at portto. "With our crypto wallet solution, users can easily start transacting crypto assets without much prior knowledge of the underlying technology."

When portto was developing Blocto in 2019, the team realized that it needed a public cloud platform that supports flexible scalability to deploy its crypto wallet, because the market was new and dynamic. Meanwhile, as a crypto wallet provider, it needed to guarantee its users and partners a high level of data security. After reviewing several offerings, the company chose to deploy its blockchain services on Google Cloud (https://cloud.google.com/) for its high scalability around the world and comprehensive information security measures.

"The blockchain market has been growing quickly but sometimes with big fluctuations. To ensure smooth user experiences and leverage resources in the most cost-effective way, we needed to run our crypto wallet with a cloud infrastructure that can be easily scaled up and down at any time," notes Lee. "Google Cloud offers excellent infrastructure scalability with proven high security, which is also essential for our services."

Supporting high scalability and low latency around the world

With the timely technical support provided by Google Cloud and its partner CloudMile (https://mile.cloud/), portto successfully launched Blocto on Google Cloud in 2019. The company uses Kubernetes clusters hosted by Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) to support its crypto wallet and virtual machines (VMs) on Compute Engine (https://cloud.google.com/compute) to run services that require less scalability. Because of the booming popularity of cryptocurrencies and NFTs, portto's crypto wallet has experienced up to 70% monthly user growth in the past years. By leveraging the highly automated scalability of GKE, the company is able to seamlessly tackle any traffic increases and ensure the constant availability of its services.

"With GKE, we can scale up and down computing resources according to the current needs at ease. Our crypto wallet service can therefore run without glitches, even when facing a 10X instant traffic surge following a product launch."

—Hao Chang, VP Engineering, portto

"The dynamics of the blockchain market are very unpredictable, so we have to be always prepared for any demand fluctuations. With GKE, we can scale up and down computing resources according to the current needs at ease," notes Hao Chang, VP Engineering at portto. "Our crypto wallet service can therefore run without glitches, even when facing a 10X instant traffic surge following a product launch."

The widely distributed data centers of Google Cloud have also helped portto provide better services to users around the world. The company stores NFT images and videos bought and sold via Blocto in Cloud Storage (https://cloud.google.com/storage) in the regions that are closer to its end users, which shortens file access latency and optimizes the user experiences that it offers.

Enabling smooth and cost-effective blockchain data indexing with Cloud Bigtable

One of the key features of the blockchain technology is its ability to protect data against manipulation by storing data in blocks that are linked together through cryptography. As the number of transactions processed through blockchains increases, the amount of data and the complexity of blockchain networks grow. Chang says that this poses technical challenges to a crypto wallet provider like portto, because the raw data stored in blockchains are increasingly hard to read or query without further processing. For example, when Blocto's users request their transaction or asset records, it could take around 20 seconds for portto's system to identify and retrieve the data from blockchains, which compromises the user experience.

To address this, portto uses remote procedure call (RPC) nodes, which are a type of server that enables access to blockchain data, to retrieve raw data from the blockchains linked with Blocto, cache the data on Cloud Bigtable (https://cloud.google.com/bigtable) for indexing, and then store the indexed data in Cloud SQL (https://cloud.google.com/sql) along with all user data of its crypto wallet app. With the indexed data, Blocto's users can see their transaction and asset records displayed on the app in less than one second.

Blocto crypto wallet

According to Chang, using Cloud Bigtable to cache the blockchain data retrieved through RPC nodes helps reduce costs and ensure sufficient database scalability. By caching the data, portto can reduce its usage of PRC nodes during the indexing process, which translates into lower operational costs. In addition, the nearly limitless automatic scalability of the NoSQL databases on Cloud Bigtable assures that portto is always able to effortlessly process the ever-increasing blockchain data.

"Indexing blockchain data is an important part of our operations, because it makes it easier to query and manage all the transaction and asset data on blockchains," he adds. "Cloud Bigtable enables us to process huge amounts of data at ease by providing much higher scalability than SQL databases. Without the automated scalability feature of Cloud Bigtable, it would have taken us one to two months to build a similar capability on SQL databases."

Guaranteeing high security with IAM and Cloud Key Management

For portto, another major advantage of adopting Google Cloud is strengthened security. Besides benefiting from the fact that the infrastructure of Google Cloud is certified by several international standards for information security like ISO 27001, portto also uses Identity and Access Management (https://cloud.google.com/iam) (IAM) to implement strict access control within the company. To guarantee transaction security on its crypto wallet platform, portto leverages Cloud Key Management (https://cloud.google.com/security-key-management) to encrypt the private keys that its users use for digital signing.

Portto's team

"Users' and partners' trust is the foundation of our business as a crypto wallet provider, and Google Cloud provides us with various tools to easily ensure high security in every aspect of our operations," notes Chang. "For example, without Cloud Key Management, we would have needed a large IT team and enormous resources to realize the same level of security for key encryption."

Further enhancing security and streamlining data pipeline with Google Cloud

portto is always working on improving its services and operations. Moving forward, it will leverage Confidential VM (https://cloud.google.com/compute/confidential-vm/docs/about-cvm) to build an all-in-one digital signing system that integrates various blockchain signing algorithms, so that the private keys used for digital signing are always protected even at the moment that they are decrypted for use. For more efficient security monitoring, portto will adopt Security Command Center (https://cloud.google.com/security-command-center) to realize centralized security and risk management.

"By leveraging the highly secure and scalable infrastructure of Google Cloud, we're able to offer an easy-to-access crypto wallet in a cost-effective manner. We believe that we can continue upgrading our product by taking full advantage of the high-performance tools of Google Cloud."

—Hsuan Lee, Co-Founder and CEO, portto

Furthermore, to establish a more streamlined data pipeline, the company plans to move its indexed blockchain data from Cloud SQL to Cloud Bigtable, which will also support data visualization and analytics. It will also make its crypto wallet product available on Google Cloud Marketplace (https://cloud.google.com/marketplace) to reach more customers.

"By leveraging the highly secure and scalable infrastructure of Google Cloud, we're able to offer an easy-to-access crypto wallet in a cost-effective manner," says Lee. "We believe that we can continue upgrading our product by taking full advantage of the high-performance tools of Google Cloud."